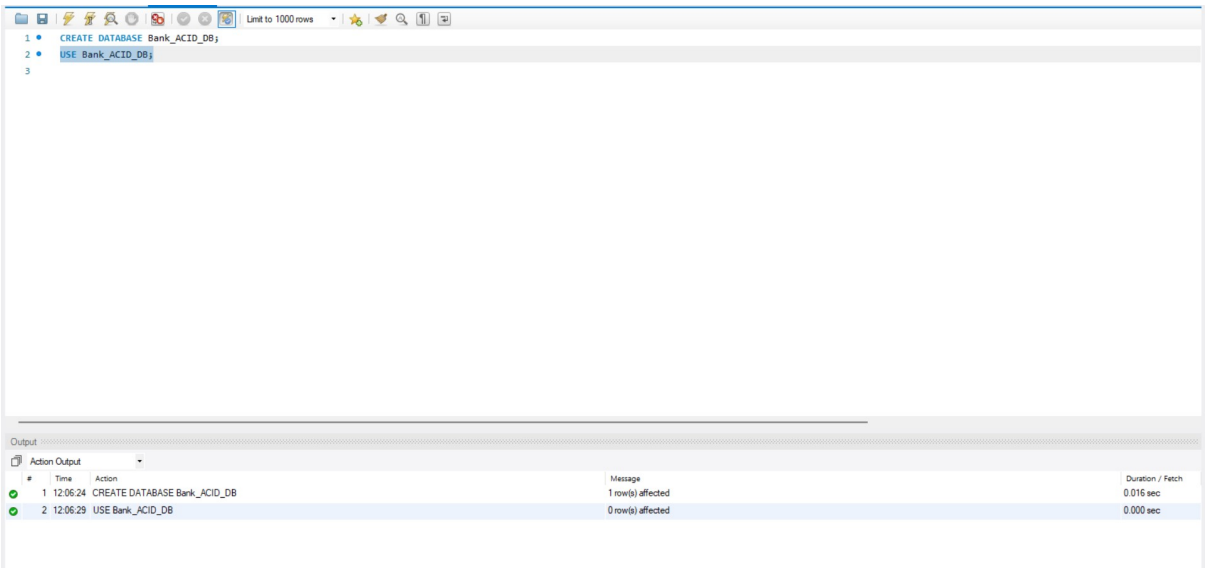
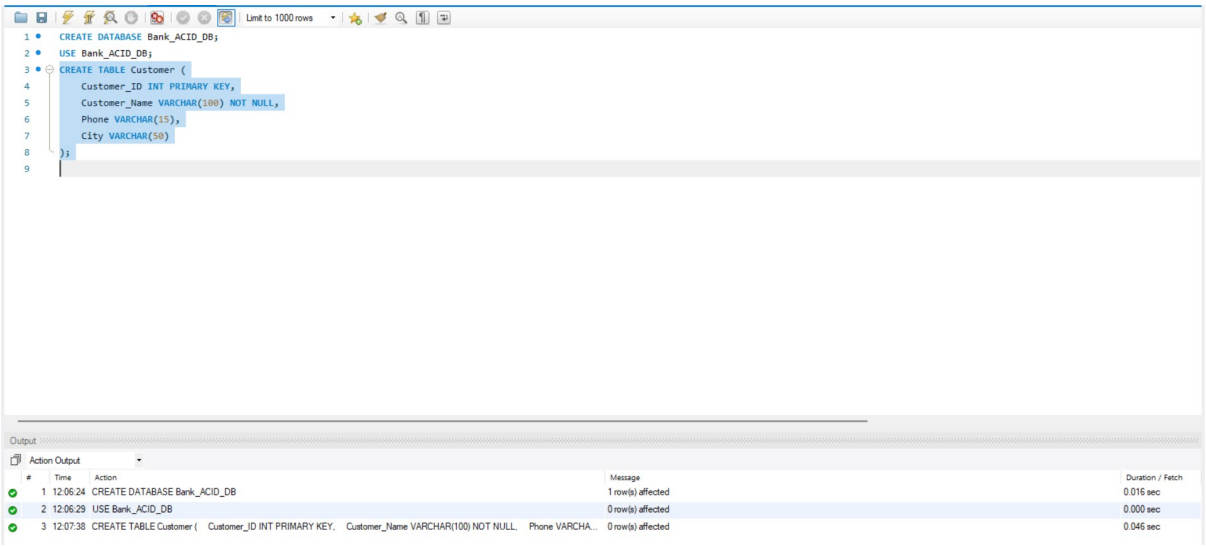
ACID PROPERTIES & ISOLATION

1. CREATE DATABASE



2. CREATE CUSTOMER TABLE



3. CREATE ACCOUNT TABLE

```
);
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),

    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:06:24	CREATE DATABASE Bank_ACID_DB	1 row(s) affected	0.016 sec
2	12:06:29	USE Bank_ACID_DB	0 row(s) affected	0.000 sec
3	12:07:38	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.046 sec
4	12:09:58	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.031 sec

4. CREATE TRANSACTION TABLE

```
CREATE TABLE Bank_Transaction (
    Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
    Account_No INT,
    Transaction_Type VARCHAR(20),
    Amount DECIMAL(12,2),
    Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,

    FOREIGN KEY (Account_No)
    REFERENCES Account(Account_No)
);
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:06:24	CREATE DATABASE Bank_ACID_DB	1 row(s) affected	0.016 sec
2	12:06:29	USE Bank_ACID_DB	0 row(s) affected	0.000 sec
3	12:07:38	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.046 sec
4	12:09:58	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.031 sec
5	12:10:42	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction...	0 row(s) affected	0.031 sec

5. INSERT CUSTOMER DATA

```
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'poojitha sai', '9876543210', 'Hyderabad'),
(102, 'priya', '9876543211', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),
(104, 'geethika', '9876543213', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	12:06:24	CREATE DATABASE Bank_ACID_DB	1 row(s) affected	0.016 sec
2	12:06:29	USE Bank_ACID_DB	0 row(s) affected	0.000 sec
3	12:07:38	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCHA...	0 row(s) affected	0.046 sec
4	12:09:58	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance DE...	0 row(s) affected	0.031 sec
5	12:10:42	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transaction...	0 row(s) affected	0.031 sec
6	12:12:12	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'poojitha sai', '9876543210', 'Hyderabad'), (102, 'p...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec

6. INSERT ACCOUNT DATA

```
INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'poojitha sai', '9876543210', 'Hyderabad'),
(102, 'priya', '9876543211', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),
(104, 'geethika', '9876543213', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	12:06:29	USE Bank_ACID_DB	0 row(s) affected	0.000 sec
3	12:07:38	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL, Phone VARCH...	0 row(s) affected	0.046 sec
4	12:09:58	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance D...	0 row(s) affected	0.031 sec
5	12:10:42	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transactio...	0 row(s) affected	0.031 sec
6	12:12:12	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'poojitha sai', '9876543210', 'Hyderabad'), (102, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
7	12:13:18	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 103, 'Savings', 50000, 'Hyde...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`bank_acid_db`.`account`, CONSTRAINT `account...	0.015 sec
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyde...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec

7. INSERT BANK TRANSACTION DATA

```
37
38 • INSERT INTO Bank_Transaction
39   (Account_No, Transaction_Type, Amount)
40   VALUES
41   (10001, 'DEPOSIT', 10000),
42   (10002, 'DEPOSIT', 15000),
43   (10003, 'WITHDRAW', 20000),
44   (10004, 'DEPOSIT', 5000),
45   (10005, 'WITHDRAW', 10000);
46
47
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	12:07:38	CREATE TABLE Customer (Customer_ID INT PRIMARY KEY, Customer_Name VARCHAR(100) NOT NULL Phone VARCH...	0 row(s) affected	0.046 sec
4	12:09:58	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance D...	0 row(s) affected	0.031 sec
5	12:10:42	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transactio...	0 row(s) affected	0.031 sec
6	12:12:12	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'poojitha sai', '9876543210', 'Hyderabad'), (102, ...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
7	12:13:18	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 103, 'Savings', 50000, 'Hyde...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`bank_acid_db`.`account`, CONSTRAINT `account...	0.015 sec
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyde...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOSI...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec

8. DISPLAY DATA

```
45     (10005, 'WITHDRAW', 10000);
46
47 • SELECT * FROM Customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Customer_ID	Customer_Name	Phone	City
101	poojitha sai	9876543210	Hyderabad
102	priya	9876543211	Vijayawada
103	Arjun Reddy	9876543212	Bangalore
104	geethika	9876543213	Chennai
105	Kiran Kumar	9876543214	Hyderabad
NULL	NULL	NULL	NULL

Customer 1 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	12:09:58	CREATE TABLE Account (Account_No INT PRIMARY KEY, Customer_ID INT, Account_Type VARCHAR(20), Balance ...	0 row(s) affected	0.031 sec
5	12:10:42	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transacti...	0 row(s) affected	0.031 sec
6	12:12:12	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'poojitha sai', '9876543210', 'Hyderabad'), (102...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
7	12:13:18	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 103, 'Savings', 50000, 'Hyd...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`bank_acid_db`.`account`, CONSTRAINT `accoun...	0.015 sec
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyd...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec

```
49 • SELECT * FROM Account;
50
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Account 2 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	12:10:42	CREATE TABLE Bank_Transaction (Transaction_ID INT PRIMARY KEY AUTO_INCREMENT, Account_No INT, Transacti...	0 row(s) affected	0.031 sec
6	12:12:12	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'poojitha sai', '9876543210', 'Hyderabad'), (102...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
7	12:13:18	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 103, 'Savings', 50000, 'Hyd...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`bank_acid_db`.`account`, CONSTRAINT `accoun...	0.015 sec
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyd...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
11	12:18:08	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

```
51 • SELECT * FROM Bank_Transaction;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-27 12:15:44
2	10002	DEPOSIT	15000.00	2026-08-27 12:15:44
3	10003	WITHDRAW	20000.00	2026-08-27 12:15:44
4	10004	DEPOSIT	5000.00	2026-08-27 12:15:44
5	10005	WITHDRAW	10000.00	2026-08-27 12:15:44
NULL	NULL	NULL	NULL	NULL

Bank_Transaction 3 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	12:12:12	INSERT INTO Customer (Customer_ID, Customer_Name, Phone, City) VALUES (101, 'poojitha sai', '9876543210', 'Hyderabad'), (102...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.000 sec
7	12:13:18	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 103, 'Savings', 50000, 'Hyd...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`bank_acid_db`.`account`, CONSTRAINT `accoun...	0.015 sec
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyd...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
11	12:18:08	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	12:18:28	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec

9. START A TRANSACTION

```
START TRANSACTION;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	12:13:18	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 103, 'Savings', 50000, 'Hyd...	Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`bank_acid_db`.`account`, CONSTRAINT `accoun...	0.015 sec
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyd...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
11	12:18:08	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	12:18:28	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	12:19:14	START TRANSACTION	0 row(s) affected	0.000 sec

10. DEPOSIT MONEY

```
START TRANSACTION;
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	12:14:45	INSERT INTO Account (Account_No, Customer_ID, Account_Type, Balance, Branch) VALUES (10001, 101, 'Savings', 50000, 'Hyd...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.016 sec
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
11	12:18:08	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	12:18:28	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	12:19:14	START TRANSACTION	0 row(s) affected	0.000 sec
14	12:20:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec

11. CHECK BALANCE

```
SELECT * FROM Bank_Transaction;

START TRANSACTION;
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
SELECT *
FROM Account
WHERE Account_No = 10001;
```

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	55000.00	Hyderabad

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	12:15:44	INSERT INTO Bank_Transaction (Account_No, Transaction_Type, Amount) VALUES (10001, 'DEPOSIT', 10000), (10002, 'DEPOS...	5 row(s) affected Records: 5 Duplicates: 0 Warnings: 0	0.015 sec
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
11	12:18:08	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	12:18:28	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	12:19:14	START TRANSACTION	0 row(s) affected	0.000 sec
14	12:20:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	12:20:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

12. COMMIT TRANSACTION

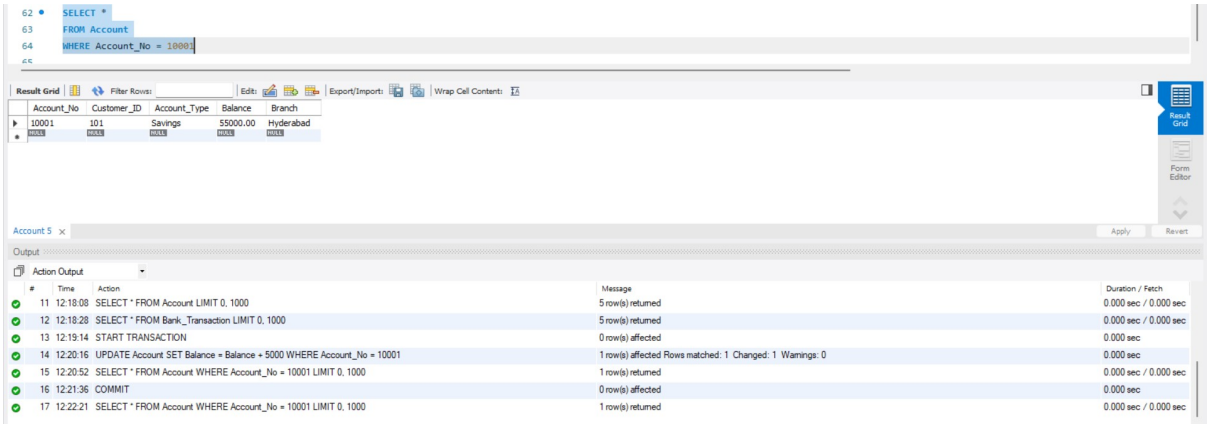
```
WHERE Account_No = 10001;
COMMIT;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	12:16:51	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned	0.015 sec / 0.000 sec
11	12:18:08	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
12	12:18:28	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned	0.000 sec / 0.000 sec
13	12:19:14	START TRANSACTION	0 row(s) affected	0.000 sec
14	12:20:16	UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001	1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0	0.000 sec
15	12:20:52	SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
16	12:21:36	COMMIT	0 row(s) affected	0.000 sec

13. CHECK BALANCE AFTER COMMIT


SELECT *
FROM Account
WHERE Account_No = 10001;
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
Account_No
Customer_ID
Account_Type
Balance
Branch
10001
101
Savings
55000.00
Hyderabad
Result Grid
Form Editor
Account 5
Apply
Revert
Output
Action Output
Time Action Message Duration / Fetch
11 12:18:08 SELECT * FROM Account LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
12 12:18:28 SELECT * FROM Bank_Transaction LIMIT 0, 1000 5 row(s) returned 0.000 sec / 0.000 sec
13 12:19:14 START TRANSACTION 0 row(s) affected 0.000 sec
14 12:20:16 UPDATE Account SET Balance = Balance + 5000 WHERE Account_No = 10001 1 row(s) affected Rows matched: 1 Changed: 1 Warnings: 0 0.000 sec
15 12:20:52 SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
16 12:21:36 COMMIT 0 row(s) affected 0.000 sec
17 12:22:21 SELECT * FROM Account WHERE Account_No = 10001 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec